

# Unit 15 - Chapter 19 - Study Guide

## Key Concepts

### 19.1 Acid-Base Theories

- Acids taste sour, bases taste bitter and feel slippery. Both are electrolytes and cause indicators to change colors.
- In an aqueous solution, an **Arrhenius acid** yields hydrogen ions and an **Arrhenius base** yields hydroxide ions.
- A **Brønsted-Lowry acid** is a hydrogen-ion donor; a **Brønsted-Lowry base** is a hydrogen-ion acceptor.
- A **Lewis acid** is an electron-pair acceptor; a **Lewis base** is an electron-pair donor.

### 19.2 Hydrogen Ions and Acidity

- For an aqueous solution, the product of  $[H^+]$  and  $[OH^-]$  equals  $1.0 \times 10^{-14}$
- On the pH scale, 0 is strongly acidic, 7 is neutral, and 14 is strongly basic.
- The acid and base form of an indicator have different colors in solution.

### 19.3 Strengths of Acids and Bases

- The stronger an acid is, the larger its  $K_a$  value.
- To find  $K_a$  of a weak acid or  $K_b$  of a weak base, substitute the concentrations of the substances into the equilibrium expression.

### 19.4 Neutralization Reactions

- An acid and a base react to produce salt and **water**.
- The point of neutralization is the **end point** of a titration.

### 19.5 Salts in Solution

- Salts that produce acidic solutions contain positive ions that release protons to water; salts that produce basic solutions have negative ions that attract protons from water.
- A **buffer** is a solution of a weak acid or weak base and one of its salts.

## Key Equations:

Ion-product constant for water:  $K_w = [H^+] \times [OH^-] = 1.0 \times 10^{-14}$

pH Formula:  $pH = -\log[H^+]$

pOH Formula:  $pOH = -\log[OH^-]$

Acid Dissociation Constant:  $K_a = \frac{[H^+][A^-]}{[HA]}$

**Key Vocab:**

- acid dissociation constant ( $K_a$ )
- acidic solution
- alkaline solution
- amphoteric
- base dissociation constant ( $K_b$ )
- basic solution
- buffers
- buffer capacity
- conjugate acid
- conjugate acid-base pair
- conjugate base
- diprotic acid
- end point
- equivalence point
- hydronium ion ( $H_3O^+$ )
- ion-product constant for water ( $K_w$ )
- Lewis acid
- Lewis base
- monoprotic acids
- neutral solution
- neutralization reaction
- pH
- salt hydrolysis
- self-ionization
- standard solution
- strong acid
- strong base
- titration
- triprotic acid
- weak acid
- weak base